KRTAP8-1 (KAP8.1, UniProt Q8IUC2): Functional Annotation Report

Summary

KRTAP8-1 encodes keratin-associated protein 8.1 (KAP8.1), a small structural protein of the hair fibre. It is not an enzyme, transporter, or signaling molecule. Its primary and essentially sole function is **structural**: it is one of the matrix proteins that embed and crosslink the keratin intermediate filaments (IFs) of the hard α-keratin composite that constitutes the hair/wool cortex. KAP8.1 belongs to the **high glycine–tyrosine (HGT) class** of keratin-associated proteins (KAPs), a diagnostic classification confirmed both by its amino-acid composition (computed here directly from the 63-residue Q8IUC2 sequence: ~42.9% Gly+Tyr, only 4 cysteines) and by the literature (PMID: 25279319, PMID: 23791804).

Mechanistically, the most precise experimental evidence comes from in-vitro biophysical work showing that **KAP8.1 binds the head domain of the type II trichocyte (hair) keratin K85** and helps organize the helical arrangement of intermediate filaments in the hair **orthocortex** — the compartment distributed asymmetrically on the convex side of curved fibres (PMID: 23791804). Through this binding, and through additional bidirectional interactions with other partners, KAP8.1 contributes to the mechanical strength and the shape/curvature of the hair fibre. Immunohistochemistry confirms that the protein is deposited in discrete regions of the keratinizing cortex, consistent with a role in controlling hair curvature (PMID: 30125694).

Structurally, AlphaFold modelling performed in this investigation shows KAP8.1 to be **fully intrinsically disordered** (mean pLDDT 41.2; 100% of residues below pLDDT 50), consistent with its role as a flexible component of the amorphous interfilamentous matrix rather than a globular folded protein. Its glycine-tyrosine core is **conserved across mammals under purifying selection**, indicating a specific, maintained keratin-interaction function. It acts **intracellularly** within terminally differentiating cortical trichocytes, is encoded by an **intronless gene** in the chromosome 21q22.1 KAP gene cluster, and is expressed early in the follicle-differentiation program under **HOXC13** control (PMID: 15385554, PMID: 22044071).

Gene-identity verification: The gene symbol KRTAP8-1 matches the UniProt protein description (Keratin-associated protein 8-1 / high glycine-tyrosine keratin-associated protein 8.1), the organism is correctly *Homo sapiens*, and the protein family (KRTAP type 8, HGT-KAP,

Pfam PF11759) aligns with all literature found. There was **no gene-symbol ambiguity**: all retrieved literature (KAP8/KAP8.1 in human, sheep, and mouse) refers to the same orthologous HGT-KAP. This report therefore proceeds with confidence on the correct target.

Key Findings

Finding 1 — KAP8.1 is a high glycine-tyrosine keratin-associated structural matrix protein of the hair cortex

The amino-acid composition of KAP8.1, computed directly from the UniProt Q8IUC2 sequence (63 residues), is the defining signature of the high glycine-tyrosine (HGT) KAP class: **glycine 23.8% and tyrosine 19.0%, together 42.9%** of all residues, with very low cysteine content (6.3%, 4 residues). The sequence contains a defined region of twelve 2-amino-acid G-[YCGS] repeats spanning residues 12–54. This composition contrasts sharply with the other two KAP classes — high-sulfur (HS) and ultra-high-sulfur (UHS) KAPs — which are cysteine-rich and mediate disulfide crosslinking. The three-class scheme is established in the literature: keratin-associated proteins are "classified into three groups; high sulfur (HS) KAPs, ultra high sulfur (UHS) KAPs, and high glycine-tyrosine (HGT) KAPs" (PMID: 23791804). KAP8 specifically is assigned to the HGT family and "was believed to be coded for by a single gene in both humans and sheep" (PMID: 25279319).

Functionally, UniProt annotates KAP8.1 as a matrix protein that embeds the hair-keratin intermediate filaments, and it belongs to the KRTAP type 8 family (Pfam PF11759). Together these establish the fundamental identity of the protein: a **non-enzymatic, non-catalytic structural matrix protein** of the hair shaft, whose role is to be embedded among, and to crosslink, the keratin IFs.

Finding 2 — KAP8.1 binds the head domain of hair keratin K85 and organizes the helical arrangement of intermediate filaments in the orthocortex

The most precise mechanistic evidence for KAP8.1 function comes from a dedicated biophysical study (Matsunaga et al., 2013). Using **isothermal titration calorimetry (ITC), Western blotting, circular dichroism, and transmission electron microscopy**, the authors demonstrated that "KAP8.1 binds to the head domain of K85, which is helically aligned around

the axis of the intermediate filament (IF)" (PMID: 23791804). K85 is a type II trichocyte (hair) keratin, so this identifies the **specific molecular binding partner and binding mode** of KAP8.1 — a level of detail rarely available for KAP proteins.

The same study links this binding directly to higher-order fibre architecture: "the helical arrangement of IFs found in the orthocortex, which is uniquely distributed on the convex fiber side of the hair, is regulated by KAP8.1" (PMID: 23791804). The authors further describe a **bidirectional binding property** of HGT-KAPs (also engaging, for example, the desmoplakin C-terminus), proposing that this crosslinking contributes to the mechanical strength and the shape of the hair. This is the central mechanistic claim of the report: KAP8.1 is a **molecular organizer/crosslinker** that positions and bundles keratin IFs on one side of the fibre.

Finding 3 — KAP8.1 localizes to discrete regions of the keratinizing cortex and is linked to hair curvature

Monoclonal-antibody immunohistochemistry (Akiba et al., 2018) showed that "KAP8.1 was predominantly expressed in discrete regions of the keratinizing zone of the hair shaft cortex" (PMID: 30125694), paralleling the distribution of KAP8.1 mRNA. The authors explicitly "suggested that KAP8.1 plays a role associated with cells to control hair curvature" (PMID: 30125694). UniProt's tissue-specificity annotation is consistent: expression is essentially restricted to only one vertical half of the hair-forming compartment, and in beard hairs is absent from the central medulla.

This asymmetric deposition is the hallmark of HGT-KAPs generally: proteins "high in glycine and tyrosine tend to be expressed initially on the orthocortical side of the follicle" (PMID: 17697142). The asymmetry is mechanistically important because curved (crimped) hairs have a bilateral cortex — ortho- on the convex side and para- on the concave side — and differential matrix-protein deposition between the two halves generates the differential swelling/mechanical behaviour that bends the fibre. KAP8.1's orthocortical localization therefore ties its molecular binding function (Finding 2) directly to the macroscopic phenotype of **hair curvature**.

Finding 4 — KRTAP8-1 is an intronless hair-cortex gene in the HGT-KAP program, co-expressed early under HOXC13 control

KRTAP8-1 maps to the **chromosome 21q22.1 KAP gene domain** and, like other HGT-KAPs, is a **single-exon (intronless) gene** — a common architecture in the KAP superfamily. Its temporal and spatial expression follows the stereotyped HGT-KAP program. In the orthologous mouse

Krtap16 HGT-KAP complex, the genes show "strong expression in a narrowly defined pattern restricted to the lower and middle cortical region of the hair shaft" (PMID: 15385554), decline through catagen, and carry functional **Hoxc13 binding sites**, placing the HGT-KAP program under the control of this master hair-follicle transcription factor.

HGT-KAP genes are also among the first KAP genes activated during follicle differentiation: "The genes encoding the high glycine-tyrosine (HGT) KAPs are the first sub-group of KAP genes expressed in the wool follicle and just after expression of the keratin genes" (PMID: 22044071). This temporal ordering — keratins first, then HGT-KAPs — is mechanistically sensible: the IF scaffold must be laid down before (or while) the matrix proteins that embed it are deposited.

Finding 5 — KAP8.1 is predicted to be fully intrinsically disordered with regularly spaced aromatic residues

An AlphaFold model (AF-Q8IUC2-F1, v6) analyzed in this investigation has a **mean pLDDT of 41.2, with 100% of residues below pLDDT 50** — the signature of an intrinsically disordered protein with no confidently folded globular structure. This is entirely consistent with KAP8.1's role as a flexible component of the **amorphous interfilamentous matrix** rather than a rigid folded module.

The sequence features that drive both disorder and function are striking: the 63-aa chain is **28.6% aromatic** (18 Tyr/Phe/Trp residues, spaced roughly every 3–4 residues), **23.8% glycine** (15 Gly, similarly regularly spaced), with only 4 cysteines (positions 3, 14, 29, 49) and a near-neutral net charge (+1). Seven GY and four YG dipeptide motifs recur. The regular spacing of aromatic residues is important mechanistically: because KAP8.1 is cysteine-poor, it cannot rely on the disulfide-crosslinking chemistry used by high-sulfur KAPs; instead, **aromatic (Tyr/Phe) stacking and other non-covalent interactions** are the likely basis of its keratin adhesion. This fits the broader biomaterial model in which "hard α-keratins are fibre-reinforced biomaterials that consist of 10 nm intermediate filaments (IFs) embedded in an elastomeric protein matrix" (PMID: 23135675) — KAP8.1 being a component of the elastomeric matrix phase.

Finding 6 — KAP8.1 has a highly conserved central core across mammals, indicating purifying selection on its keratin-interaction function

Retrieval and alignment of 25 mammalian KAP8.1/KRTAP8-1 orthologs from UniProt (human Q8IUC2, sheep P02448, mouse O08633, cow F1MTA9, orangutan H2P2Y0, and others) revealed a **strongly conserved central block**. Measured against human as the anchor, core identity was: **orangutan 100%, cow 93%, sheep 91%, mouse 74%** over the ~42–44-aa core. A central motif

around **PLGYSVGCGYGSTYSPVGYG** — including the GY-rich repeat and conserved Cys/Ser/Pro residues — is near-invariant, whereas the N-terminal ~8–9 residues are more variable. The retention of this core under purifying selection implies that it mediates a **specific, functionally important keratin interaction** that natural selection has preserved for tens of millions of years.

This bioinformatic conclusion is independently corroborated by experiment: the Akiba et al. monoclonal antibody "recognized a linear epitope shared among human, mouse, and sheep KAP8.1" (PMID: 30125694) — i.e., a conserved linear stretch of the protein across exactly the species in which high core identity was measured here.

Mechanistic Model / Interpretation

KAP8.1 can be understood as a small, disordered **molecular "glue and organizer"** operating inside terminally differentiating hair-cortex cells. The following model synthesizes the six findings.

The hard α-keratin composite

Hair is a fibre-reinforced biological composite. Its building block is the macrofibril: paracrystalline arrays of ~10 nm **keratin intermediate filaments (IFs)** embedded in an **amorphous protein matrix** made of KAPs (PMID: 23135675). KAP8.1 is a matrix protein in this composite.

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Hair cortex composite (cross-section, schematic)

||||  ||||  ||||      <- keratin intermediate filaments (incl. K85)
||||  ||||  ||||
::::::::::::::::      <- amorphous KAP matrix
::KAP8.1::KAP8.1::       (HGT, cysteine-poor, aromatic/Gly-rich)
::::::::::::::::

KAP8.1 binds the HEAD DOMAIN of K85, which is helically
arrayed around the IF axis -> organizes helical IF bundling
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From molecule to fibre shape

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Gene (chr 21q22.1, intronless)
       |  HOXC13-driven, early in differentiation
       v
KAP8.1 protein (63 aa, intrinsically disordered)
       |  Gly 23.8% + Tyr 19.0%; aromatic-rich; Cys-poor
       v
Binds head domain of hair keratin K85  -- bidirectional crosslinking
       |
       v
Organizes helical arrangement of IFs in the ORTHOCORTEX
       |  deposited asymmetrically (convex side of fibre)
       v
Bilateral ortho/para cortex asymmetry
       |
       v
Mechanical strength + hair SHAPE / CURVATURE
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The key mechanistic links are:

Level	What KAP8.1 does	Evidence
Sequence	Gly/Tyr-rich, Cys-poor, aromatic — adhesion via non-covalent (aromatic) chemistry	Composition computed from Q8IUC2; PMID: 25279319
Structure	Intrinsically disordered → flexible matrix filler	AlphaFold pLDDT 41.2, 100% < 50
Molecular interaction	Binds head domain of keratin K85	ITC/CD/TEM, PMID: 23791804
Supramolecular	Organizes helical IF arrangement in orthocortex	PMID: 23791804
Tissue	Deposited in discrete keratinizing cortex regions, one vertical half	PMID: 30125694, PMID: 17697142
Phenotype	Contributes to mechanical strength and hair curvature	PMID: 23791804, PMID: 30125694
Evolution	Core conserved under purifying selection → specific maintained function	Ortholog alignment; PMID: 30125694

Localization

KAP8.1 carries out its function **intracellularly**, within the cytoplasm of terminally differentiating cortical trichocytes of the hair follicle, where it is deposited into the hardening keratin matrix during keratinization. It is not secreted and is not a membrane protein. Within the fibre, it is spatially restricted to the **orthocortex** (one vertical half of the cortex) and is absent from the medulla in beard hairs (PMID: 30125694). After keratinization is complete, the protein persists as a permanent, crosslinked structural constituent of the dead hair shaft.

Pathway context

KAP8.1 does not participate in a classical signaling or metabolic pathway. The relevant "pathway" is the **hair-follicle keratinization / differentiation program**: hair keratin genes are expressed first, immediately followed by the HGT-KAP subgroup (including KRTAP8-1), under the transcriptional control of HOXC13 (PMID: 22044071, PMID: 15385554). The output of this program is the assembled hard α-keratin fibre.

Evidence Base

PMID	Title (abbreviated)	How it supports the findings
23791804	*Bidirectional binding property of high glycine-tyrosine keratin-associated protein contributes to the mechanical strength and shape of hair*	**Primary mechanistic study.** ITC/CD/TEM show KAP8.1 binds the head domain of keratin K85 and organizes helical IF arrangement in the orthocortex; establishes HGT-KAP class and bidirectional crosslinking → strength and shape. Supports Findings 1, 2.
30125694	*Structural behavior of keratin-associated protein 8.1 in human hair as revealed by a monoclonal antibody*	**Localization study.** IHC shows KAP8.1 in discrete keratinizing cortex regions; proposes role in hair curvature; demonstrates a conserved linear epitope across human, mouse, sheep. Supports Findings 3, 6.
25279319	*The sheep KAP8-2 gene, a new KAP8 family member that is absent in humans*	Assigns KAP8 to the HGT-KAP family; notes single-gene status in humans. Supports Finding 1.
15385554	*Krtap16 … hair keratin-associated protein (KAP) gene complex … regulation by Hoxc13*	Cortical, stage-specific expression of the HGT-KAP program; functional HOXC13 binding sites. Supports Finding 4.
22044071	*Search for variation in the ovine KAP7-1 and KAP8-1 genes …*	Places HGT-KAPs first in the KAP expression sequence, just after hair keratins. Supports Finding 4.
17697142	*The differential expression of proteins in the cortical cells of wool and hair fibres*	HGT proteins are deposited initially on the orthocortical side → basis for crimp/curvature asymmetry. Supports Finding 3.
23135675	*Regulation of hard α-keratin mechanics via control of intermediate filament hydration: matrix squeeze revisited*	Establishes the IF-in-matrix composite architecture in which KAP8.1 is the elastomeric matrix phase. Supports Finding 5.

Supporting context papers from the broader literature set reinforce the model without directly testing KAP8.1: characterization of human type I/II hair keratin and KAP gene domains (PMID: 14962103, PMID: 15617563, PMID: 15737194); the human KAP review (PMID: 16939781); mouse

HGT proteins (PMID: 9374545); and hair mechanical/viscoelastic studies (PMID: 28919511, PMID: 22848616, PMID: 27162354) that describe the IF/matrix composite whose matrix phase KAP8.1 populates.

Limitations and Knowledge Gaps

1. **No KAP8.1-specific loss-of-function phenotype in humans.** There is no reported human disease directly attributable to KRTAP8-1 mutation. KAP genes are highly polymorphic and partially redundant, so a single-gene knockout may be phenotypically subtle. The curvature/strength role is therefore inferred from biochemistry, localization, and orthology rather than from a human genetic lesion.
2. **Binding studied for K85 only.** The direct binding evidence is for the head domain of keratin K85 (PMID: 23791804). Whether KAP8.1 binds additional hair keratins (e.g., other type II trichocyte keratins) with comparable affinity, and the precise residues mediating binding, remain to be mapped.
3. **The aromatic-adhesion hypothesis is inferential.** The proposal that Tyr/Phe stacking (rather than disulfide chemistry) mediates keratin adhesion is well-motivated by composition and disorder but has not been directly demonstrated by mutagenesis for KAP8.1.
4. **AlphaFold confidence is intrinsically low for disordered proteins.** The pLDDT < 50 result is itself evidence of disorder, but it means no atomic-resolution structural model of the bound state exists. The conformation KAP8.1 adopts upon binding keratin is unknown.
5. **Quantitative contribution to fibre mechanics is unresolved.** The degree to which KAP8.1 specifically (versus the HGT-KAP class collectively) sets curvature or strength has not been isolated experimentally.
6. **Human vs. ovine/murine extrapolation.** Several mechanistic details derive from sheep and mouse systems; while the core is highly conserved (Finding 6), species-specific differences in expression timing or partner keratins cannot be excluded.

Proposed Follow-up Experiments / Actions

1. **Targeted binding map.** Perform ITC/SPR of recombinant human KAP8.1 against a panel of hair keratin head domains (K81–K86 type II; K31–K40 type I) to define substrate specificity beyond K85.
2. **Mutagenesis of aromatic residues.** Systematically substitute the conserved Tyr/Phe residues (and the near-invariant core motif PLGYSVGCGYGSTYSPVGYG) and re-test keratin binding to directly test the aromatic-adhesion hypothesis.
3. **Knockout/knock-in mouse.** Generate Krtap8-1 null and point-mutant mice and quantify hair curvature, cross-sectional ortho/para asymmetry, and tensile/viscoelastic properties to isolate KAP8.1's phenotypic contribution.
4. **Super-resolution immuno-EM.** Localize KAP8.1 at the IF surface within orthocortical macrofibrils to test the predicted helical, IF-head-domain-associated positioning in situ.
5. **Population-genetics of KRTAP8-1 variants.** Associate human KRTAP8-1 polymorphisms with quantitative hair-curl phenotypes across ancestries to test the curvature link in humans.
6. **Integrative structural modelling.** Use AlphaFold-Multimer / disordered-protein ensemble methods on the KAP8.1–K85 head-domain complex to generate a testable bound-state model.

Report generated from a 5-iteration autonomous investigation: 6 confirmed findings, 35 papers reviewed. Target gene identity (KRTAP8-1 / Q8IUC2 / Homo sapiens / HGT-KAP type 8 family) was verified against all literature with no gene-symbol ambiguity.
